

# **Towards a semantic structure for classifying IoT agriculture sensor datasets : An approach based on machine learning and web semantic technologies**

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## ***ABSTRACT ORIGINAL***

With the increase in the number of IoT farming datasets, it has become so difficult to identify the right data for IoT agriculture applications. Therefore, a meaningful structure is needed to well understand, interpret and index IoT farming datasets. This paper proposes a new IoT farming ontology that allows the organization, the understanding, and the classification of IoT agriculture datasets knowledge as well as meta-data storage. For this, we have developed a new IoT agriculture taxonomy that helps to identify an IoT agriculture application based on the combination of various IoT agriculture sensors. The evaluation of the semantic IoT agriculture datasets classification, based on the background knowledge provided by the proposed ontology, was achieved using Machine Learning algorithms, including Logistic Regression, Decision Tree Classifier, K-Neighbors Classifier, Linear Discriminant Analysis, Gaussian NB, SVM, and Random Forest Regressor. The obtained results clearly show the effectiveness of the proposed ontology to classify IoT agriculture datasets with high performances and accuracy (0.98), (0.99) using Decision tree classifier and SVM respectively. © 2023 The Author(s)